

Conformal Mappings

Sections 9.1–9.2

Complex Analysis

What is a Conformal Mapping?

A function

$$w = f(z)$$

is called **conformal** at a point z_0 if it preserves angles and orientation between curves passing through z_0 .

In complex analysis:

f is conformal at z_0

if

- ▶ f is analytic at z_0
- ▶ $f'(z_0) \neq 0$

Local Behavior of Analytic Functions

Let f be analytic near z_0 .

Then

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + O((z - z_0)^2).$$

For small $|z - z_0|$,

$$w - f(z_0) \approx f'(z_0)(z - z_0).$$

Thus locally the mapping behaves like

- ▶ a **rotation**
- ▶ combined with a **scaling**

Angle Preservation

Let two curves pass through z_0 with tangents

$$\frac{dz_1}{dt}, \quad \frac{dz_2}{dt}.$$

Under the mapping

$$w = f(z)$$

their images have tangents

$$\frac{dw}{dt} = f'(z_0) \frac{dz}{dt}.$$

Thus

$$\arg \left(\frac{dw_1}{dw_2} \right) = \arg \left(\frac{dz_1}{dz_2} \right).$$

Therefore analytic functions with $f'(z_0) \neq 0$ preserve angles.

Definition

A mapping $w = f(z)$ is **conformal** in a region if

- ▶ f is analytic in the region
- ▶ $f'(z) \neq 0$ everywhere in the region

Examples:

- ▶ $f(z) = z^2$
- ▶ $f(z) = e^z$
- ▶ $f(z) = \sin z$

These mappings are conformal except where $f'(z) = 0$.

Example

Consider

$$f(z) = z^2.$$

Derivative:

$$f'(z) = 2z.$$

Thus the mapping is conformal everywhere except

$$z = 0.$$

Near any other point the mapping locally behaves like

$$w \approx 2z_0(z - z_0).$$

Section 9.2: Bilinear Transformations

A **bilinear transformation** (or Möbius transformation) has the form

$$w = \frac{az + b}{cz + d}$$

where

$$ad - bc \neq 0.$$

These mappings play a fundamental role in conformal mapping theory.

Basic Properties

For

$$w = \frac{az + b}{cz + d}$$

the derivative is

$$\frac{dw}{dz} = \frac{ad - bc}{(cz + d)^2}.$$

Thus

$$f'(z) \neq 0$$

whenever $cz + d \neq 0$.

Therefore bilinear transformations are conformal everywhere except at

$$z = -\frac{d}{c}.$$

Inverse Transformation

A bilinear transformation is invertible.

Solving

$$w = \frac{az + b}{cz + d}$$

for z gives another bilinear transformation:

$$z = \frac{dw - b}{-cw + a}.$$

Thus the inverse of a Möbius transformation is also Möbius.

Mapping of Circles and Lines

A fundamental property:

Bilinear transformations map **circles and lines** to **circles or lines**.

In the extended complex plane, lines are regarded as circles through ∞ .

Important Special Transformations

Every bilinear transformation can be built from:

1. translations

$$w = z + b$$

2. rotations and scalings

$$w = az$$

3. inversion

$$w = \frac{1}{z}$$

These elementary maps help analyze the geometry of Möbius transformations.

Mapping Three Points

A bilinear transformation is uniquely determined by the images of three distinct points.

Given

$$z_1, z_2, z_3$$

and

$$w_1, w_2, w_3$$

there exists a unique Möbius transformation satisfying

$$f(z_k) = w_k.$$

Summary

Section 9.1:

- ▶ conformal mappings preserve angles
- ▶ analytic functions with $f'(z) \neq 0$ are conformal
- ▶ locally analytic maps behave like rotations and scalings

Section 9.2:

- ▶ bilinear transformations
- ▶ mapping of circles and lines
- ▶ determined by three points